

A COVERING FOR COLLAPSE: TUBES, SHEETS, CLUSTERS, AND THE REDUCTION TO LIOUVILLE PROBLEMS

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ABSTRACT. We organise the ways a first-time Navier–Stokes singularity could concentrate vorticity, and show that, modulo stated hypotheses, each ends either in an exclusion by the mechanisms of the companion papers (leak, drift, supply) or in a Liouville-type problem. The key new estimate bounds the strain at a collapsing core due to vorticity beyond distance r by $C\sqrt{A}/r^2$, with A the scale-invariant local energy; since compression needs $\gamma \geq 2\nu/r_c^2$, the strain of any collapse comes from within $C'(A/\nu^2)^{1/4}$ core radii, whatever the number or sign of distant structures. With a dichotomy for weak- L^3 mass outside the backward paraboloid, this sends spread concentration to Seregin’s weighted framework and sparse concentration (satellite tubes) to near clusters. A torn tube’s own sheet does not re-protect; sheets pooled from several tubes re-form a core only below a pooled threshold $\varepsilon_r \approx 0.03$, and the re-formed whole is then carried off by the drift. The remaining problems are the Navier–Stokes Liouville problem for self-stretching clusters (whose members, by the Terminus, must be protected or symmetric), an Euler Liouville problem, and a discretely self-similar one.

1. INTRODUCTION

A singularity of the Navier–Stokes equations must concentrate vorticity, and the companion papers describe what happens to one kind of concentration, a slender vortex tube in an external strain: in the tearing regime it leaks circulation at an exponentially small but scale-invariant rate [12], its centroid drifts off the stagnation point at a structure-independent rate [13], and, if it survives on its own circulation, it must gather circulation at a rate that requires growing local energy [14]. This paper asks whether every concentration falls under these mechanisms, and organises the alternatives.

Section 2 states when the tube reduction is valid. Section 3 proves that the strain driving a collapsing core comes from within a bounded number of core radii, whatever lies further away. Section 4 sorts concentrations into tubes, sheets and clusters; Section 5 shows that torn sheets do not re-protect, and that pooled sheets re-form a core only below a threshold. Section 6 treats concentrations fed from outside the backward paraboloid, which split into spread energy (Seregin’s weighted class) and sparse satellite tubes. Section 7 proves that an isolated tearing tube cannot be an ancient solution and derives geometric constraints on protected members, and Section 8 reduces the cluster problem to excluding a self-regenerating cluster of protected, curved, expanding loops. Section 9 lists what remains: three Liouville-type problems, for Navier–Stokes, for Euler, and for discretely self-similar profiles.

2. VALIDITY OF THE TUBE REDUCTION (T-A)

Proposition 2.1 (slender tube in a slowly varying strain). *Let a vortex tube have core radius a , axial variation length L_z , curvature radius R_c , and lie in an external field whose nearest sources are at distance $\geq d$. In the frame translating with the core and with $D(t) = \nabla u_{\text{ext}}$ at*

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the centroid, the axial vorticity obeys the reduction $\partial_t \omega + \nabla \cdot ((Dx + u_\omega)\omega) = \nu \Delta \omega + R$ with $\|R\| \lesssim \omega_0 s [(a/L_z)^2 + a/R_c + a/d]$.

Proof sketch. The axial and curvature terms are the classical slender-vortex asymptotics (Callegari–Ting; Ting–Klein): at leading order the cross-section obeys the 2D equation with axial stretching, the curvature entering at $O(a/R_c)$ through the binormal (local-induction) motion and a dipolar core correction. The strain non-uniformity follows from the far-strain lemma with one more derivative: integrating by parts against a smooth cutoff, $|\nabla^2 u_{\text{far}}(0)| \leq C\sqrt{A}/r^3$, so D varies across the core by $\lesssim (a/d)|D|$. $\square$

3. THE FAR-STRAIN LEMMA

Let u be a finite-energy velocity field with vorticity ω , place the centroid of a core at 0, and let $A(R) = R^{-1} \int_{B_R} |u|^2$. Let $\chi_r(y) = \chi(|y|/r)$ with χ smooth, $\chi = 0$ on $[0, 1]$, $\chi = 1$ on $[2, \infty)$, so $|\nabla \chi_r| \leq C/r$. The velocity gradient at 0 induced by the vorticity at distance $\gtrsim r$ is

$$\nabla u_{\text{far}}(0) = \int \nabla K(-y) \chi_r(y) \omega(y) dy, \quad |\nabla K(y)| \leq C|y|^{-3}, \quad |\nabla^2 K(y)| \leq C|y|^{-4},$$

with K the Biot–Savart kernel, and S_{far} is its symmetric part.

Lemma 3.1 (far strain is weak). $|S_{\text{far}}| \leq |\nabla u_{\text{far}}(0)| \leq C\sqrt{A_{\text{sup}}}/r^2$, where $A_{\text{sup}} = \sup_{R \geq r} A(R)$ and C is absolute.

Proof. Write $\omega = \nabla \times u$ and integrate by parts once, moving the derivative onto $\chi_r \nabla K(\cdot)$. To justify this, insert a second cutoff $1 - \chi_\rho$ with $\rho \rightarrow \infty$; its contribution is bounded like term (b) below at radius ρ , i.e. by $C\sqrt{A(2\rho)}/\rho^2 \leq C\sqrt{2E_0}\rho^{-5/2} \rightarrow 0$, since $A(R) \leq 2E_0/R$ for a finite-energy field. There are no other boundary terms, because χ_r vanishes near the singularity of K . So $|\nabla u_{\text{far}}(0)| \leq (a) + (b)$ with

$$(a) = C \int_{|y| > r} |y|^{-4} |u| dy, \quad (b) = C \int_{r < |y| < 2r} |\nabla \chi_r| |y|^{-3} |u| dy.$$

(a) On the shell $S_k = \{2^k r < |y| < 2^{k+1} r\}$, by Cauchy–Schwarz,

$$\int_{S_k} |y|^{-4} |u| \leq (2^k r)^{-4} |S_k|^{1/2} \|u\|_{L^2(B_{2^{k+1}r})} \leq C(2^k r)^{-4} (2^k r)^{3/2} (2^{k+1} r A(2^{k+1} r))^{1/2} = C\sqrt{A(2^{k+1} r)} (2^k r)^{-2},$$

and summing over $k \geq 0$ gives $(a) \leq C\sqrt{A_{\text{sup}}}/r^2$. (b) Similarly, $(b) \leq Cr^{-1}r^{-3}|B_{2r}|^{1/2}\|u\|_{L^2(B_{2r})} \leq Cr^{-4}r^{3/2}(2rA(2r))^{1/2} = C\sqrt{A(2r)}/r^2$. $\square$

The proof uses only $\|u\|_{L^2}$ on shells: it is independent of the number, coherence and sign of the distant structures, and no cancellation between them is needed.

Corollary 3.2 (strain suppliers are near). *A core of mean-square radius r_c can be compressed only by an external compression $\gamma \geq 2\nu/r_c^2$ [9]. Hence at least half of it comes from vorticity within $r_* = C'(A_{\text{sup}}/\nu^2)^{1/4} r_c$ of the core, with $A_{\text{sup}} = \sup_{R \geq r_*} A(R)$.*

Proof. By Lemma 3.1, vorticity beyond r supplies at most $C\sqrt{A_{\text{sup}}}/r^2$, which is at most ν/r_c^2 once $r \geq (C\sqrt{A_{\text{sup}}}/\nu)^{1/2} r_c$. $\square$

For a Type I core, $r_c \sim \sqrt{T_* - t}$ and A is bounded, so its strain suppliers lie within a fixed multiple of the parabolic radius. The compression bound of [9] is stated in the tube reduction and assumes, as throughout the tube branch, that the external strain is roughly uniform across the core.

4. THE TRICHOTOMY

(T) slender tube: Terminus (leak, drift); (H) via T-a (Proposition 2.1) and T-b (companion paper, numerical gap). (S) sheet: below. (B) cluster: Liouville core. G1 is not needed: the corollary applies to any core.

5. SHEETS (S)

Proposition 5.1 (sketch). *A torn tube's stripped sheet is extended along the outflow at rate $|g_2|$; its line density decays like $e^{-|g_2|t}$, and the strain it induces on any fragment is $O(\Gamma_{\text{sheet}}/L(t)^2)$ (an infinite flat sheet induces none), so a fragment's local $\lambda_i = \lambda + O(\Gamma e^{-2|g_2|t}/(L_0^2 s))$. Since $\Gamma_i \leq \Gamma$, $\varepsilon_i \geq \varepsilon > \varepsilon_c$ after an $O(1/|g_2|)$ transient: no fragment re-protects without supply.*

Numerically confirmed (G2-S, G2-S': six cases, no protected fragment at either threshold).

Proof. Fragments. Define fragments as connected components of a superlevel set of $\omega \geq 0$ (as in the numerics); each has circulation $\Gamma_i \leq \Gamma$ because circulation is conserved and $\omega \geq 0$, so at fixed λ_i , $\varepsilon_i = 2\pi\lambda_i\nu/\Gamma_i \geq \varepsilon$. *Strain from the sheet.* The stripped vorticity lies along the outflow axis; the material length of the layer grows as $L(t) \geq L_0 e^{|g_2|t}$, so its line density is $\gamma_s \leq \Gamma_{\text{sh}} e^{-|g_2|t}/L_0$. At a point at distance d from the layer, $|S| \leq \int |\omega(y)|/(\pi|x-y|^2) dy \leq \int \gamma_s dy/(\pi(d^2 + y^2)) + \Gamma_{\text{sh}}/(\pi L^2) \leq \gamma_s/d + \Gamma_{\text{sh}}/(\pi L^2)$. For a fragment of radius $a_f \leq d$: $|\delta\lambda_i| = 2|S|/s \leq \frac{2\Gamma_{\text{sh}}}{sL_0}(\frac{1}{a_f} + \frac{1}{L_0})e^{-|g_2|t}$. *Transient.* Protection requires $\lambda_i < 1$, i.e. $|\delta\lambda_i| > \lambda - 1$, which can hold only for $t < t_* = |g_2|^{-1} \ln(\frac{2\Gamma_{\text{sh}}(1/a_f + 1/L_0)}{sL_0(\lambda-1)})$; afterwards $\lambda_i \geq 1$ and $\varepsilon_i \geq \varepsilon > \varepsilon_c$, so no fragment re-protects without supply. $\square$

Remark 5.2. The bound is crude (an infinite flat layer induces no strain, so the true $\delta\lambda_i$ is smaller); the numerics found $\lambda_i = \lambda$ to 1% throughout. The transient t_* is logarithmic in the sheet's circulation, so (S) folds into supply after a bounded time.

Remark 5.3 (pooled sheets: the regeneration window). Proposition 5.1 concerns one parent. A layer pooled from several parents, with total circulation Γ_L along the outflow axis, was evolved in the reduction ($\lambda \in \{0.8, 1, 1.2, 1.5, 2\}$, $\Gamma_L/\nu \in \{300, 1257\}$, to $t = 30/s$). With $\varepsilon_L = 2\pi\lambda\nu/\Gamma_L$, it rolls up into a single round core holding all its circulation whenever $\varepsilon_L \leq 0.025$, including tearing strain ($\lambda = 1.2, 1.5, 2$ at $\Gamma_L = 1257$; $\lambda = 1.2$ at 300), and fails for $\varepsilon_L \geq 0.031$ (elongated remnant, swept out). The regathering threshold $\varepsilon_r \approx 0.03$ lies a little below the runaway threshold of a formed core. A regathered core is not kept: by the moment law (Proposition 5.4), in a uniform strain any configuration is carried off at $|g_2|$; in these runs only the point symmetry of the initial state pinned it.

Proposition 5.4 (moment law). *In the reduction, for ω of either sign, $\frac{d}{dt} \int x \omega dx = D \int x \omega dx$. Hence a configuration with $\Gamma \neq 0$ has its circulation centroid carried off along the outflow at $|g_2|$, and one with $\Gamma = 0$ has its first moment (for a dipole, Γd) obey $d' = Dd$ unless circulation cancels.*

Proof. $\frac{d}{dt} \int x \omega = \int (Dx + u_\omega) \omega + \nu \int x \Delta \omega$; the last term vanishes (two integrations by parts, decay), and $\int u_\omega \omega = \iint K(x-y) \omega(x) \omega(y) = 0$ because the Biot–Savart kernel is odd. $\square$

6. EXTERNAL SUPPLY: A DICHOTOMY

Let $(0, T_*)$ be a first-time singularity, $\theta(t) = \sqrt{a(T_* - t)}$ as in Barker–Fernández-Dalgo–Prange [8], and $\sigma = -\ln(T_* - t)$. Their hypothesis (H_{out}) is a bound on $\|v(t)\|_{L^{3,\infty}}$ on the annulus $\theta < |x| < \sqrt{a}$; when it fails, we ask whether the local energy must grow.

Lemma 6.1 (dichotomy). *Fix t with σ large, and let $K = \|v(t)\|_{L^{3,\infty}(\theta < |x| < \sqrt{a})}$, $K_{\text{eff}} = K/(2(\sigma/\ln 2)^{1/3})$. Then there are a level $\lambda > 0$ and a dyadic shell $D_j = \{|v| > \lambda\} \cap \{\theta < |x| < \sqrt{a}\} \cap \{2^j \leq |x| < 2^{j+1}\}$ with*

$$A(2^{j+1}) \geq \frac{1}{2} K_{\text{eff}}^2 f_j^{1/3}, \quad f_j = |D_j|/8^j$$

(f_j is, up to a constant, the fraction of shell j that D_j fills).

Proof. (1) By the definition of the weak- L^3 quasi-norm, $K = \sup_\lambda \lambda |E_\lambda|^{1/3}$ with $E_\lambda = \{|v| > \lambda\} \cap \{\theta < |x| < \sqrt{a}\}$, so some λ has $\lambda |E_\lambda|^{1/3} \geq K/2$. (2) The annulus is covered by $J \leq \log_2(\sqrt{a}/\theta) + 1 = \sigma/(2 \ln 2) + 1 \leq \sigma/\ln 2$ dyadic shells, so by pigeonhole some shell has $|D_j| \geq |E_\lambda|/J$, hence $\lambda |D_j|^{1/3} \geq \frac{K}{2} J^{-1/3} \geq K_{\text{eff}}$. (3) Then

$$A(2^{j+1}) \geq \frac{\lambda^2 |D_j|}{2^{j+1}} = \frac{(\lambda |D_j|^{1/3})^2 |D_j|^{1/3}}{2^{j+1}} \geq \frac{K_{\text{eff}}^2 f_j^{1/3} 2^j}{2^{j+1}} = \frac{1}{2} K_{\text{eff}}^2 f_j^{1/3}. \quad \square$$

Corollary 6.2 (spread or sparse). *Suppose (H_{out}) fails, i.e. $K(t_n) \rightarrow \infty$ along a sequence $t_n \rightarrow T_*$, and fix $A_* > 0$. At each t_n , with j and λ from Lemma 6.1, either (I) spread: $A(2^{j+1}) > A_*$ at a scale $2^{j+1} \in (\theta, 2\sqrt{a})$; or (II) sparse: $f_j \leq 8A_*^3/K_{\text{eff}}^6$, while $\lambda |D_j|^{1/3} \geq K_{\text{eff}}$.*

Proof. If $A(2^{j+1}) \leq A_*$, Lemma 6.1 gives $f_j^{1/3} \leq 2A_*/K_{\text{eff}}^2$. $\square$

Since K_{eff} loses only a factor $\sigma^{1/3}$, a failure of (H_{out}) faster than $\sigma^{1/3}$ gives either unbounded local energy at intermediate scales, which is Seregin's weighted class [7], or intense structures with vanishing filling fraction outside the paraboloid. Thin intense structures are tubes: a tube of circulation Γ_s , radius a_s and length ℓ contributes $\sim \Gamma_s a_s^{-1/3} \ell^{1/3}$ to the weak- L^3 norm, large exactly when thin. So branch (II) is supply by satellite tubes arriving from outside the paraboloid. By Corollary 3.2, satellites at distances $r \gg \sqrt{T_* - t}$ from a Type I core supply no leading-order strain: they are passengers, and the core's strain comes from a near cluster. An infinite hierarchy of satellites is the classical supercritical chain; the Terminus applies to each level but does not exclude the chain as a whole.

7. LIOUVILLE PROBLEMS AUGMENTED BY THE TERMINUS

Theorem 7.1 (no isolated tearing tube in an ancient solution). *Let $\lambda > 1$ and let $\omega \geq 0$, with $\Gamma = \int \omega > 0$ and finite second moments, be an ancient solution of $\partial_t \omega + \nabla \cdot ((Dx + u_\omega)\omega) = \nu \Delta \omega$ on $(-\infty, 0]$. This is impossible if (a) $\Gamma < 8\pi\nu$, or (b) the second moments are bounded as $t \rightarrow -\infty$.*

Proof. (a) With y the outflow axis and $M_{yy} = \int y^2 \omega$: $\frac{d}{dt} M_{yy} = 2|g_2| M_{yy} + 2 \int \omega y u_y + 2\nu \Gamma$, and symmetrising the Biot–Savart double integral gives $2 \int \omega y u_y = T/2\pi$, $T = \iint \omega \omega r_1 r_2 / |r|^2$. Since $|r_1 r_2|/|r|^2 \leq \frac{1}{2}$ and $\omega \geq 0$, $T \geq -\Gamma^2/2$, so $\frac{d}{dt} M_{yy} \geq 2|g_2| M_{yy} + c_0$, $c_0 = 2\nu\Gamma - \Gamma^2/4\pi > 0$. Integrating on $[t, 0]$: $M_{yy}(t) \leq e^{2|g_2|t} [M_{yy}(0) + c_0/2|g_2|] - c_0/2|g_2| \rightarrow -c_0/2|g_2| < 0$ as $t \rightarrow -\infty$, contradicting $M_{yy} \geq 0$. (b) (Sketch.) Bounded moments, $\omega \geq 0$ and fixed Γ make the backward orbit tight, so its α -limit set is a non-empty compact set of eternal solutions with the same moment bound. For an eternal solution the drift $b = Dx + u_\omega$ is bounded on compact sets; by the Stroock–Varadhan support theorem, particles reach the strain-dominated outflow region ($r \gg \sqrt{\Gamma/|g_2|}$) with positive probability per unit time, uniformly on the compact α -limit set, and from there escape along y with probability $\geq \frac{1}{2}$. The escaped mass contributes $\gtrsim q\Gamma R^2 e^{2|g_2|t}$ to M_{yy} , which diverges, contradicting the moment bound. (No coherence hypothesis is needed: positivity of the leak comes from the support theorem.) $\square$

Corollary 7.2. *In any rescaled ancient limit, a tearing-regime tube cannot be isolated in a uniform strain: it belongs to a cluster with non-uniform, time-dependent mutual strain, or was fed from infinity in the past (unbounded backward moments). With Lemma 3.1, every branch flows to clusters.*

Proposition 7.3 (one straight neighbour always tears). *A straight vortex line induces a planar strain $\sigma \text{diag}(1, -1, 0)$ (zero along its own axis). A tube whose axis lies in the plane of the extension at angle α to it sees $\lambda = (1 + \sin^2 \alpha) / \cos^2 \alpha \geq 1$, with equality only when aligned. Hence a protected cluster member ($\lambda < 1$) needs at least two non-coplanar neighbour strains combining into a uniaxial extension, e.g. $\text{diag}(\sigma, -\sigma, 0) + \text{diag}(\sigma, 0, -\sigma) = \text{diag}(2\sigma, -\sigma, -\sigma)$.*

Proof. With axis $\hat{a} = (\cos \alpha, 0, \sin \alpha)$: $s = \hat{a} \cdot S\hat{a} = \sigma \cos^2 \alpha$; in the plane $\perp \hat{a}$ (basis $e_2, (-\sin \alpha, 0, \cos \alpha)$) the restricted strain is $\text{diag}(-\sigma, \sigma \sin^2 \alpha)$, so $2e = \sigma(1 + \sin^2 \alpha)$ and $\lambda = 2e/s$. $\square$

The augmented cluster problem. A bounded ancient solution (KNSS) whose concentrated vorticity sits in coherent tubes must keep every member protected or exactly symmetric (else leak and drift, Theorem 7.1), and protection requires at least two non-coplanar neighbours per member (Proposition 7.3). This is a structural restriction to rich three-dimensional geometries (coaxial rings, orthogonal arrays), not yet a Liouville theorem. *Euler limit:* with $\nu = 0$, $c_0 = -\Gamma^2/4\pi < 0$ and the leak vanishes, so Theorem 7.1(a) is unavailable; (b)'s support argument needs diffusion. The drift survives, but an Euler version is open. *DSS:* a periodic Leray orbit containing a tearing tube has an unstable Floquet direction (codimension ≥ 1), not excluded.

8. THE CLUSTER PROBLEM

Lemma 8.1 (straight tubes are not stretched on average). *For any velocity field decaying at infinity along a straight line with direction $\hat{a}$, $\hat{a} \cdot S\hat{a} = \partial_s(u \cdot \hat{a})$ along the line, so $\int_{-\infty}^{\infty} \hat{a} \cdot S\hat{a} ds = [u \cdot \hat{a}]_{-\infty}^{+\infty} = 0$. In particular two skew straight vortex lines produce zero mean axial stretching on each other (and zero stretching at the point of closest approach, where the separation is normal to both). For a closed material loop, $\frac{dL}{dt} = -\oint \kappa(u \cdot \hat{n}) ds$.*

Proposition 8.2 (thin and fat clusters). *(i) Thin clusters (core radius much smaller than separations and curvature radii) satisfy the filament approximation; there, interacting filaments cannot stretch singularly (Hormoz–Brenner), and a tearing member leaks (companion paper). (ii) Fat clusters at bounded circulation Reynolds number $R_i \sim 10$ –100 (paper IV) have $\varepsilon = 2\pi\lambda/R \gtrsim 0.06 > \varepsilon_c$, so a tearing member is torn within $O(1/s)$: every coherent member is protected ($\lambda < 1$), needs at least two non-coplanar neighbour strains (Proposition 7.3), and is a curved, expanding loop (Lemma 8.1).*

Proposition 8.3 (finite age). *In an ancient solution, a coherent member stretched at mean rate $\geq s_{\min}$ has $L(t) \leq L(0)e^{-s_{\min}|t|}$ backward, while its Burgers core keeps a $\gtrsim \sqrt{4\nu/s_{\max}}$; it is a tube only for $|t| \leq \ln(L(0)/a)/s_{\min}$. Hence an ancient cluster must regenerate its members infinitely often.*

Remark 8.4 (the iterated cascade). The regeneration cycle has prior art: Brenner–Hormoz–Pumir [10] propose, for Euler, an iteration in which a filament pair flattens into sheets that break up by Kelvin–Helmholtz into same-sign filaments, $\Gamma_{n+1} = C(b_n/a_n)\Gamma_n$, with unstable fixed points; McKeown et al. [11] observe three generations in Navier–Stokes (core size $\times 3$ smaller, a 2:1 circulation split), halted by viscosity. In that self-similar ansatz $\tau_n = R_n^2/\Gamma_n$, so $\Gamma_{n+1}/\Gamma_n = \Lambda_X^2/\Lambda_T$, and Leray's bound $\|u\|_{\infty} \geq c\sqrt{\nu/(T-t)}$ at generation n reads $\Gamma_n/\nu \geq c^2$: a

Navier–Stokes cascade needs summable per-generation losses. Splitting gives a ratio below 1, and full regathering (Remark above) still loses the leak, so summability forces $\Gamma_n/\nu \gtrsim (2\pi\lambda/c) \ln n$: gathering. Their measured exponents ($b \propto (T-t)$, $a \propto (T-t)^{-1/3}$) put the colliding cores at $\lambda = 2$, so the flattening is tearing, which in Navier–Stokes requires $\varepsilon > \varepsilon_c$ (in Euler it is inevitable). This holds inside the ansatz only.

Status. The cluster problem reduces to excluding a *self-regenerating* cluster of protected, curved, expanding loops, each with at least two non-coplanar neighbours and each of finite age: the eternal bounded cycle that the KNSS conjecture rules out (proved in two dimensions and in axisymmetric cases). The Terminus does not close it but removes straight, tearing, eternal and isolated members.

9. WHAT REMAINS

Layer 1: Navier–Stokes KNSS (clusters, whose tearing-regime members leak and drift, so they must be protected or symmetric); Euler, bounded scale-invariant energy (gathering, slowly varying weights); DSS (recycling). Layer 2: T-b proof; Proposition 5.1 rigorous; Lorentz details; BFP localisation.

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